

Notes: Ellison (JPE, 2002)

Evolving Standards for Academic Publishing: A q-r Theory

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1 Big picture

- **Question.** Why have academic papers become longer, more reference-heavy, and more revision-intensive, and why has the review process slowed down?
- **Core mechanism.** The paper models quality as two-dimensional: q is the importance of the core idea, while r is polish, generality, robustness, literature integration, extensions, and other features often improved during revisions.
- **Observed trend.** Many disciplines, especially economics, experienced longer papers, more references, and longer review/revision delays from the 1970s to the late 1990s.
- **Static insight.** Different social norms about how to weight q versus r can all be self-consistent.
- **Dynamic insight.** A learning model in which referees infer standards from their own experiences and from papers they referee can generate gradual movement toward greater emphasis on r -quality.
- **Economic takeaway.** The slowdown of academic publishing may reflect endogenous social norms that increasingly reward polish, robustness, and revision work over raw idea quality.

2 Incremental contribution of the paper

- The paper creates a formal q - r language for publication standards, separating idea quality from polish and revision quality.
- It links descriptive facts about longer papers and longer review delays to an equilibrium model of academic standards.
- It shows that standards can be self-fulfilling: many q/r weighting norms can be consistent with rational author and referee behavior.
- It provides a dynamic mechanism in which biased social learning can produce a slow drift toward r -quality.
- It contributes beyond congestion stories by treating publication delay and paper form as endogenous consequences of social norms.

3 Data and measurement

3.1 Table 1: Mean References per Nonreview Article and Average Page Lengths

The paper compares average page lengths in 1975 and 1999 and average references in 1977/78 and 1998 across leading journals in many disciplines. Economics journals show large increases: *Econometrica* rises from 14.8 to 31.1 pages, *QJE* from 15.5 to 36.7, *JPE* from 22.2 to 29.7, and *REStud* from 12.9 to 23.5.

Interpretation. These variables proxy for r -quality: exposition, literature positioning, robustness, and extensions.

3.2 Table 2: Duration of the Review Process at Various Journals

The paper compares review-process duration in 1975 and 1999. *Econometrica* rises from 9.3 to 26.3 months, *Journal of Finance* from 6.5 to 18.6 months, and *REStud* from 12.1 to 28.8 months.

Interpretation. Review delay is another proxy for r -quality because revisions typically add robustness, exposition, and extensions.

4 Models and theoretical specifications

4.1 Static q-r model

There is a continuum of academics and one journal that publishes mass τ of papers. Papers have two qualities, q and r . A social norm accepts papers satisfying

$$\alpha q + (1 - \alpha)r \geq z.$$

Authors first choose $t_q \in [0, 1]$ and draw $q \sim F(q | t_q)$. Referees assess q and ask for an r -quality threshold

$$r(q) = \frac{z - \alpha q}{1 - \alpha}.$$

Authors then choose revision time $t_r \in [0, 1 - t_q]$ and produce

$$r = h(t_r) + \eta, \quad \eta \sim U[0, \sigma].$$

Editors accept the top τ papers.

Interpretation. The timing assumption is central: revisions improve r , not q . This creates a trade-off between developing ideas and polishing papers.

4.2 Revision decision and equilibrium effort

Very low- q papers are hopeless and not revised. Intermediate- q papers are revised fully but may still fail. High- q papers require only enough revision to clear the threshold. The first-stage effort solves

$$t_q^* \in \arg \max_t G^{-1}(1 - \tau; t),$$

where $G(z; t)$ is the probability that overall quality is below z when idea effort is t and revisions are chosen optimally.

Interpretation. The model rationalizes immediate rejections, uncertain revise-and-resubmits, and easy acceptances within a single framework.

4.3 Consistent social norms

For each $\alpha \in [0, 1]$, there exists a unique $z^*(\alpha)$ such that $(\alpha, z^*(\alpha))$ is a consistent social norm.

Interpretation. The static model has a continuum of self-consistent standards. This multiplicity is why dynamic learning can matter.

4.4 Dynamic model of evolving norms

Academics believe the norm is (α_t, z_t) , write papers, referee papers, infer a new norm $(\hat{\alpha}_t, \hat{z}_t)$, and update according to

$$(\alpha_{t+1}, z_{t+1}) = (\alpha_t, z_t) + k[(\hat{\alpha}_t, \hat{z}_t) - (\alpha_t, z_t)].$$

Without overconfidence, beliefs return quickly near the equilibrium locus. With small overconfidence, academics hold others to standards that are slightly too high, observe surprise acceptances of heavily revised papers, and infer that r is more important than previously believed.

Interpretation. Overconfidence plus a continuum of norms creates a slow drift toward lower α and greater weight on r -quality.

5 Identification and theoretical strategy

- The paper is a theoretical and descriptive exercise, not a causal empirical design.
- Step 1 documents facts: longer papers, more references, and longer review delays.
- Step 2 interprets these facts as increases in r -quality.
- Step 3 builds a static equilibrium model in which many q/r norms are self-consistent.
- Step 4 uses comparative statics to discuss technologies, idea exhaustion, knowledge accumulation, and selectivity.
- Step 5 builds a dynamic learning model to explain gradual drift in standards.
- Main limitation: other forces, such as congestion and professional growth, may also explain some trends.

6 Tables and figures

6.1 Table 1: Mean References per Nonreview Article and Average Page Lengths

Read: Economics papers become much longer and more reference-heavy; similar trends appear across many fields, though magnitudes differ.

Interpretation. The table establishes the product-side change to be explained: academic articles increasingly contain observable components of r -quality.

Economic message. The rise in length and references is consistent with a profession increasingly valuing completeness and polish.

TABLE 1
MEAN REFERENCES PER NONREVIEW ARTICLE AND AVERAGE PAGE LENGTHS

FIELD	JOURNAL	PAGES		REFERENCES	
		1975	1999	1977/78	1998
Accounting	<i>Accounting Rev.</i>	12.1	20.9	12.3	32.4
Anthropology	<i>Current Anthropology</i>	11.2	17.8	61.4	37.3
Anthropology	<i>Amer. J. Physical Anthropology</i>	8.9	14.0	26.6	47.3
Biology	<i>Cell</i>	9.1	11.5	35.4	49.0
Chemistry	<i>J. American Chemical Soc.</i>	6.4	7.9	32.8	38.6
Communication	<i>J. Communication</i>	9.0	18.6	10.0	52.7
Computer science	<i>IEEE Trans. Information Theory</i>	7.3	13.7	17.5	23.9
Computer science	<i>J. ACM</i>	12.0	29.0	15.0	37.8
Computer science	<i>Communications ACM</i>	7.6	7.0	14.6	6.9
Demography	<i>Demography</i>	13.7	13.5	16.9	37.5
Ecology	<i>Ecology</i>	9.8	12.7	33.0	56.2
Economics	<i>Econometrica</i>	14.8	31.1	15.1	27.2
Economics	<i>JPE</i>	22.2	29.7	18.0	29.7
Economics	<i>QJE</i>	15.5	36.7	15.9	31.2
Economics	<i>REStud</i>	12.9	23.5	12.2	29.2
Education	<i>American Educational Res. J.</i>	15.0	36.4	19.4	50.7
Education	<i>Harvard Educational Rev.</i>	23.8	26.8	47.2	43.7
Engineering: Aerospace	<i>AIAA J.</i>	5.8	7.8	11.5	21.4
Electrical	<i>IEEE Trans. Electronic Devices</i>	6.6	7.2	13.9	17.7
Materials	<i>J. Materials Sci.</i>	7.7	7.8	18.0	21.6
Mechanical	<i>J. Engineering Mechanics and ASCE</i>	15.3	8.5	14.9	19.2
Finance	<i>J. Finance</i>	13.9	34.8	15.0	30.8
Geography	<i>Ann. Assoc. American Geographers</i>	12.7	24.4	37.1	67.1
Geology	<i>Geology</i>	4.8	4.0	19.8	25.2
Geology	<i>J. Geology</i>	17.8	15.3	35.7	50.7
History	<i>American Historical Rev.</i>	22.5	31.0	85.9	62.7
History	<i>J. American History</i>	22.8	28.4	75.2	65.9
Law	<i>Harvard Law Rev.</i>	59.1	92.4	156.6	47.7
Law	<i>Yale Law J.</i>	56.2	85.4	165.9	59.1
Linguistics	<i>Language</i>	18.8	30.1*	30.9	61.2
Mathematics	<i>Ann. Mathematics</i>	21.6	35.0	18.0	28.4
Mathematics	<i>Inventiones Mathematicae</i>	22.7	34.6	16.1	25.2
Medicine	<i>New England J. Medicine</i>	4.0	7.2	23.0	24.9
Medicine	<i>J. American Medical Assoc.</i>	4.1	6.6	9.5	24.2
Oceanography	<i>Limnology and Oceanography</i>	10.2	11.3	26.5	42.3
Paleontology	<i>J. Paleontology</i>	13.8	12.8	19.2	45.4

(a) Table 1, part 1

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TABLE 1
(Continued)

FIELD	JOURNAL	PAGES		REFERENCES	
		1975	1999	1977/78	1998
Political science	<i>American J. Polit. Sci.</i>	16.9	24.2	27.6	37.2
Political science	<i>American Polit. Sci. Rev.</i>	12.7	16.1*	37.1	51.0
Psychology	<i>Psychological Rev.</i>	21.6	30.3*	51.4	60.0
Psychology	<i>J. Personality and Soc. Psychology</i>	8.3	14.4	23.8	59.3
Psychology	<i>Developmental Psychology</i>	7.8	11.5	12.6	45.8
Sciences	<i>Nature</i>	4.2	5.7	17.8	25.8
Sciences	<i>Science</i>	7.7	5.5	23.3	28.5
Sociology	<i>American Sociological Rev.</i>	15.5	17.2	35.8	49.2
Sociology	<i>American J. Sociology</i>	22.7	39.3	26.2	60.1
Statistics	<i>J. American Statist. Assoc.</i>	5.9	11.5	13.5	28.3

(b) Table 1, continued

6.2 Table 2: Duration of the Review Process at Various Journals, 1975 and 1999

Read: Most journals take longer to process papers; economics journals show especially large increases.

Interpretation. Delay is interpreted as evidence of more revision-intensive publication standards.

Economic message. The cost of publication norms includes author, referee, and editor time.

TABLE 2
DURATION OF THE REVIEW PROCESS AT VARIOUS JOURNALS: 1975 AND 1999

FIELD	JOURNAL	MEAN DELAY	
		1975	1999
Mean Submission–Final Resubmission Time			
Biology	<i>Cell</i>	1.2	2.2
Computer science	<i>IEEE Trans. Informa- tion Theory</i>	6.9	13.6
Computer science	<i>J. ACM</i>	10.4	21.0
Economics	<i>Econometrica</i>	9.3	26.3
Engineering:			
Aerospace	<i>AIAA J.</i>	5.3	8.7
Electrical	<i>IEEE Trans. Electronic Devices</i>	4.6	6.0
Electrical	<i>Proc. IEEE</i>	2.6	6.9
Materials	<i>J. Materials Sci.</i>	1.5	5.2
Finance	<i>J. Finance</i>	6.5*	18.6
Mathematics	<i>Ann. Mathematics</i>	2.0	5.5
Mathematics	<i>Inventiones Mathematicae</i>	1.3	10.8
Philosophy	<i>Philosophy of Sci.</i>	.9	5.8
Physics	<i>Physical Rev. B</i>	.4	1.3
Political science	<i>American J. Polit. Sci.</i>	6.0	7.6
Psychology	<i>Psychological Rev.</i>	1.7	18.8
Statistics	<i>Ann. Statis.</i>	15.0	18.0
Statistics	<i>J. American Statis. Assoc.</i>	10.1	18.6
Biological	<i>Biometrika</i>	5.3	15.1
Psychological	<i>Psychometrika</i>	8.1	16.8
Mean Submission–Acceptance Time			
Ecology	<i>Ecology</i>	8.9	9.2†
Economics	<i>QJE</i>	8.5	13.0
Economics	<i>REStud</i>	12.1	28.8
Geology	<i>Geology</i>	2.2	4.6
Oceanography	<i>Limnology and Oceanography</i>	7.0	8.8‡
Mean Submission–Publication Time			
Astronomy	<i>Astrophysical J.</i>	7.9	10.7
Chemistry	<i>J. American Chemical Soc.</i>	7.8	6.2
Geology	<i>J. Geology</i>	12.4	10.6
Linguistics	<i>Language</i>	12.8	24.8
Philosophy	<i>Philosophy Soc. Sci.</i>	13.8	22.1
Physics	<i>Physical Rev. Letters</i>	2.1	5.8
Psychology	<i>Developmental Psychology</i>	9.8	24.1
Sciences	<i>Nature</i>	4.3	4.3

Table 2: Review-process duration

6.3 Figures 1–2: Time Line and Typical Equilibrium Distribution

Read: Figure 1 shows the timing of idea production, referee evaluation, revision, and editorial acceptance. Figure 2 shows the typical equilibrium regions: no revision, revised but rejected, and accepted papers.

Interpretation. The typical equilibrium formalizes the revise-and-resubmit process.

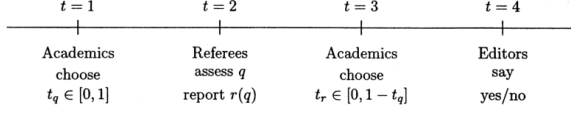


FIG. 1.—Time line

papers with $\alpha q + (1 - \alpha)r \geq z$ is exactly τ . I shall refer to (α, z) as a *consistent social norm* if there exist choices of t_r and $t_r(q)$ for which $(\alpha, z,$

(a) Figure 1: Time line

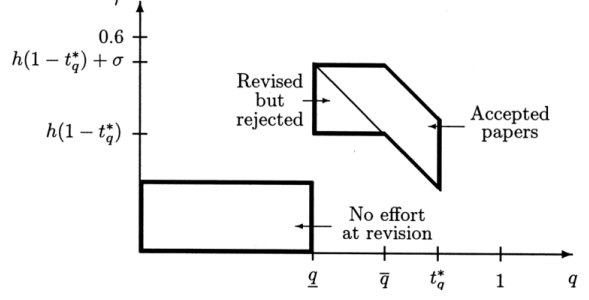


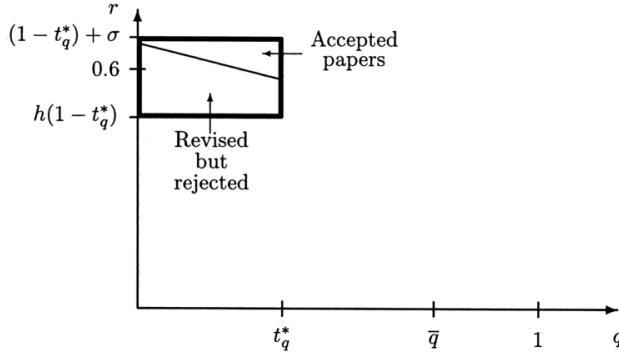
FIG. 2.—The equilibrium distribution of paper qualities in a "typical" equilibrium: $\tau = 0.3$, $q \sim U[0, t_q]$, $h(t) = \sqrt{t - (t/2)}$, $\sigma = 0.2$, $\alpha = 0.5$, and $z \approx 0.504$.

(b) Figure 2: Typical equilibrium

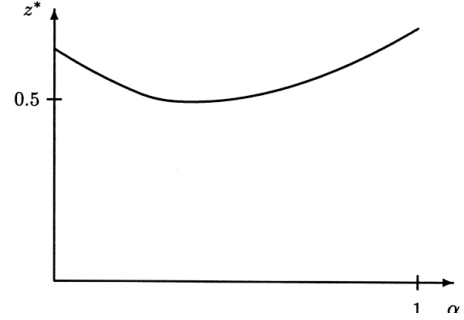
6.4 Figures 3–4: Low-Alpha Equilibrium and Consistent Social Norms

Read: Figure 3 shows a polish-heavy low- α equilibrium. Figure 4 shows the continuum of consistent norms.

Interpretation. Multiplicity of norms makes the dynamic evolution of standards central.



(a) Figure 3: Low- α equilibrium



(b) Figure 4: Consistent social norms

6.5 Figure 5: Effect of Journal Selectivity on Time Allocation and Observed Paper Quality

Read: Selectivity changes both author time allocation and the quality mix of published papers; the effect on r -quality is nonmonotonic.

Interpretation. Competition alone does not mechanically explain longer, more polished papers.

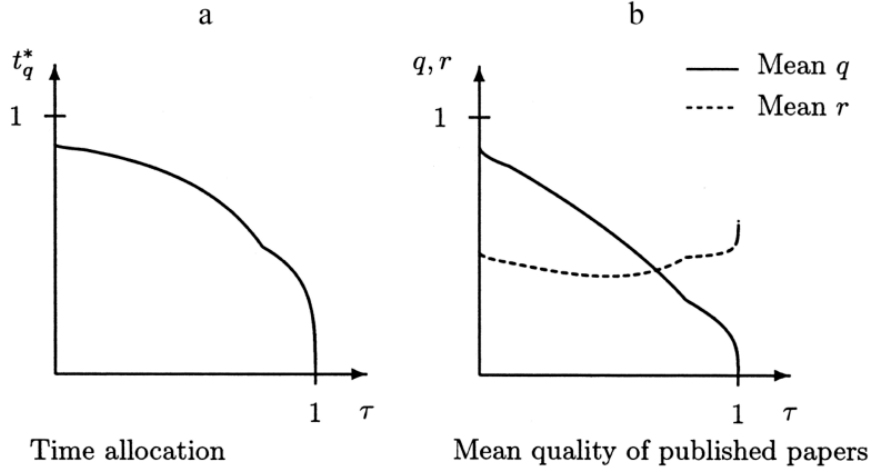
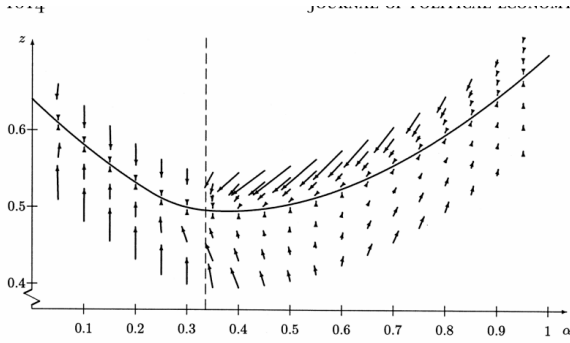


Figure 5: Journal selectivity and observed quality

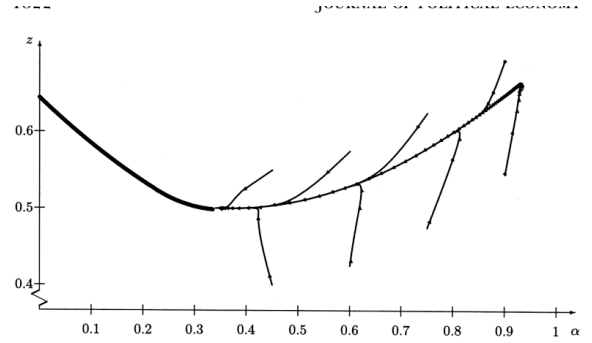
6.6 Figures 6–9: Disequilibrium Learning and Overconfidence

Read: Figure 6 shows quick convergence without bias. Figures 7 and 8 explain how too-high standards lead referees to infer more r -weight. Figure 9 shows slow drift with overconfidence.

Interpretation. These figures are the dynamic heart of the paper: small individual overconfidence can generate a long institutional movement toward r -quality.



(a) Figure 6: No-bias dynamics



(b) Figure 9: Overconfidence dynamics

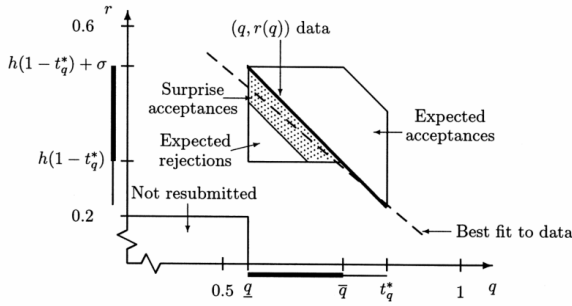
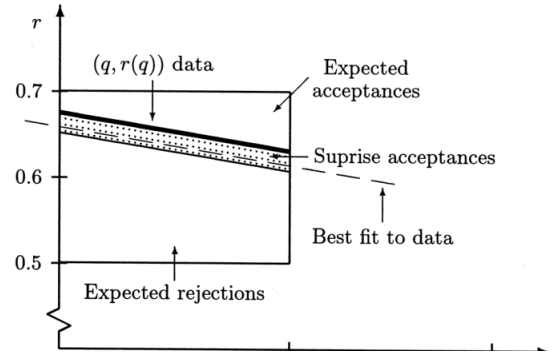


FIG. 7—Academics' inferences when standards are too high: the "typical" case

(a) Figure 7: Typical inference



(b) Figure 8: Low- α inference

7 Interpretive synthesis

The paper explains academic publishing as an equilibrium institution. Longer papers and slower review processes are not simply accidents; they can be consequences of self-confirming norms. More r -quality can improve reliability and exposition, but it can also crowd out time spent developing new ideas.

8 Short summary

- Papers have become longer, more reference-heavy, and more revision-intensive.
- The paper interprets these trends as increases in r -quality.
- A static q - r model shows that many social norms can be self-consistent.
- A dynamic model with small overconfidence generates gradual drift toward more r -quality.
- The welfare implication is ambiguous: polish may improve papers, but excessive revisions may waste effort and reduce idea production.

9 Conclusion

9.1 Core conclusion

The paper argues that the evolution of academic publishing can be understood as an endogenous shift in social norms toward higher r -quality.

9.2 Incremental contribution in one sentence

Ellison turns complaints about slower publication and longer papers into a formal theory of self-confirming and evolving academic standards.

9.3 Takeaway

If standards are partly conventional, academic communities may improve welfare by explicitly discussing how much they want to reward raw ideas versus polish and exhaustive revision.